

Data Feature-Constrained GAN Data Generation for NIR Spectral Predictive Model Calibration

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Abstract: To address the limitations in model generalization caused by insufficient sample sizes for near-infrared (NIR) spectroscopy modeling, this paper proposes a “Data Feature-Constrained Generative Adversarial Network (DFC-GAN)” method to enhance modeling reliability in small-sample scenarios through statistically constrained data generation. The method leverages the variance matrix of the top M principal components derived from PCA dimensionality reduction of the original data as a constraint, embedding it into the generator’s loss function. This ensures that the generated data maintains statistical consistency with the original data in the reduced-dimensional space while passing the discriminator’s validation. Experiments on a public dataset demonstrate that the DFC-GAN-generated calibration set data significantly improves model prediction accuracy (achieving a 17.75%, 22.85%, 50.88% reduction in RMSE for three target). The variance-matching mechanism of principal components effectively guarantees the physical interpretability and informational value of the generated data. With a clear framework, this method provides a feature-driven data augmentation solution for NIR small-sample modeling and exhibits strong potential for practical engineering applications.

Key Words: NIR, Feature-Constrained, GAN, Small Sample

1 Introduction

Near-infrared (NIR) spectroscopy is a widely used non-destructive analytical technique applied in fields such as chemical analysis, food safety, pharmaceutical testing, and environmental monitoring. Its principle is based on the light absorption characteristics of molecular substances in the near-infrared region. By measuring the absorption and reflection of near-infrared light by samples, NIR spectroscopy can provide information on the material composition of the samples. NIR technology offers advantages such as rapid analysis, non-destructiveness, and no need for sample pretreatment, making it particularly suitable for real-time analysis of complex samples. As a result, it has gained extensive application in both industrial and scientific research[1].

The core of NIR spectroscopy lies in the modeling and interpretation of spectral data. Typically, NIR spectral measurements result in a multidimensional data matrix, where each spectral point corresponds to a feature value that reflects the chemical or physical properties of the sample. By constructing appropriate mathematical models, NIR spectral data can be utilized for quantitative analysis of sample components, detection of substance concentrations, and prediction of sample quality[2]. However, due to the high dimensionality and complexity of NIR spectral data, building precise models still presents significant challenges.

Current research on NIR modeling has explored several innovative approaches. In the field of food science, NIR is frequently employed for non-contact food detection[3]. Additionally, deep learning-based NIR modeling is playing an increasingly significant role. However, since data models are highly sensitive to the scale and quality of data[4], NIR modeling becomes a complex issue that requires in-depth analysis when data quality is poor or labeled data is limited.

Scholars have already begun addressing these challenges. For instance, [5] utilized the Partial Least Squares (PLS) method for variable selection, enhancing the capability of portable NIR devices in detecting adulteration in quinoa flour. Meanwhile, [6] employed Generative Adversarial Networks (GANs) to tackle the issue of data scarcity under small sample conditions, generating synthetic training data resembling real data to enrich the calibration set and improve the detection capabilities of NIR devices.

Among these challenges, small-sample NIR modeling represents a prominent issue in current NIR spectral analysis. In practical applications, acquiring large amounts of labeled data for model training is often costly and time-consuming, particularly in fields requiring high precision where sample collection is frequently constrained. Small-sample modeling raises the critical question of how to train an efficient and accurate model with limited data while ensuring the reliability of prediction results, making it a significant challenge in NIR spectral analysis.

The challenges of small-sample NIR modeling are primarily reflected in the following aspects:

1. **Insufficient Data Diversity:** Due to the limited number of samples, the diversity of data often fails to cover all possible variations in samples. As a result, the model cannot effectively learn the characteristic information of all categories or components.
2. **High-Dimensional Data Sparsity:** NIR spectral data is typically high-dimensional, with each sample corresponding to numerous spectral features. In cases of limited sample size, this exacerbates the issues of the curse of dimensionality and overfitting.
3. **Poor Model Generalization:** With a small sample size, traditional supervised learning algorithms often suffer from severe overfitting, where the model performs well on the training set but exhibits

poor generalization in practical applications, struggling to adapt to new or unseen scenarios.

4. **Sample Imbalance:** In practical applications, the data volume of certain sample categories is significantly smaller than others, leading to sample imbalance. This not only affects modeling accuracy but may also result in the underestimation or neglect of certain categories.

Therefore, addressing the challenges posed by insufficient data through appropriate data processing, feature selection, and model optimization strategies under small-sample conditions remains a critical issue in the field of NIR modeling.

2 Overview of the Problem

This study addresses the issue of poor model identification capability caused by the lack of calibration set data under small-sample conditions. A Generative Adversarial Network (GAN) algorithm based on covariance matrix constraints derived from Principal Component Analysis (PCA) distribution is proposed, aiming to generate data with specific statistical features through the generative model. Specifically, by introducing covariance matrix constraints, the generator not only learns to generate the overall distribution of the data but also ensures that the generated data possesses a covariance structure similar to that of the training data.

2.1 Research Background and Problem Definition

In traditional Generative Adversarial Networks (GANs), the generator attempts to produce data from a noise distribution, while the discriminator tries to distinguish whether the data is real or generated. Typically, the generator learns only how to replicate the data distribution but may fail to adequately capture the statistical characteristics of the data, such as the covariance structure. To address this limitation, we introduce a novel constraint: the covariance matrix of the training data is extracted through PCA and incorporated as a constraint during the generator's training. This constraint ensures that the generated data not only resembles the real data in distribution but also maintains consistent statistical features, such as the covariance structure, with the real data.

2.2 Algorithm Goal

The main objectives of the algorithm are:

Training the Generator (G):

Enable the generator to produce data that closely resembles the real data (especially labeled data) in terms of distribution and statistical features (e.g., mean, covariance).

Training the Discriminator (D):

Enable the discriminator to distinguish between generated samples and real samples.

Covariance Constraint:

Compute the covariance matrix of the training data using PCA and incorporate it as a constraint into the generator's loss function. This ensures that the generated data maintains statistical consistency with the real data.

3 Data Feature-Constrained Generative Adversarial Network (DFC-GAN)

3.1 Data Preprocessing and PCA

We first normalize the existing small-sample calibration dataset using min-max normalization. Then, the training data is centered, as shown in the following formula (1).

$$X_{centered} = X - \mu \quad (1)$$

Subsequently, the covariance matrix is calculated in (2)

$$\Sigma = \frac{1}{n} X_{centered}^T X_{centered} \quad (2)$$

Since the principal components obtained from PCA are linearly independent, the covariance matrix of the dimensionality reduction result is a square matrix. We select the top M principal components as constraints to be introduced into the training of the generator.

3.2 GAN Training

The training of the GAN network involves two components: the generator (G) and the discriminator (D). The objective of the discriminator is to distinguish between real data and generated data by taking both as inputs. The discriminator loss function is shown in (3):

$$\mathcal{L}_D = -\mathbb{E}_{X \sim P_{real}} [\log D(x)] - \mathbb{E}_{Z \sim P_{noise}} [\log (1 - D(G(z)))] \quad (3)$$

The generator's objective is to produce data that can deceive the discriminator while maintaining statistical consistency with the real data. The generator loss function is shown in (4):

$$\mathcal{L}_G = -\mathbb{E}_{Z \sim P_{noise}} [\log D(G(z))] \quad (4)$$

Additionally, we introduce a "covariance constraint", which calculates the covariance matrix of the generated data and compares it with the target covariance matrix (derived from the training data). The total loss function for the generator is shown in (5):

$$\mathcal{L}_G = -\mathbb{E}_{Z \sim P_{noise}} [\log D(G(z))] + \lambda \|\Sigma_{gen} - \Sigma_{target}\|_F \quad (5)$$

Here, Σ_{gen} represents the covariance matrix of the generated data, Σ_{target} represents the covariance matrix of the training data, $\|\cdot\|_F$ denotes the Frobenius norm, and λ is a hyperparameter that controls the strength of the covariance constraint.

The generator and discriminator are then updated alternately. In each training epoch, the discriminator is trained using both real samples and generated samples, while the generator adjusts its output based on feedback from the discriminator. During the training process, the generator not only aims to produce high-quality data but also ensures that the generated data maintains consistency with the target covariance matrix.

The training process is illustrated in Figure 1.

GAN with PCA-based Covariance Constraint

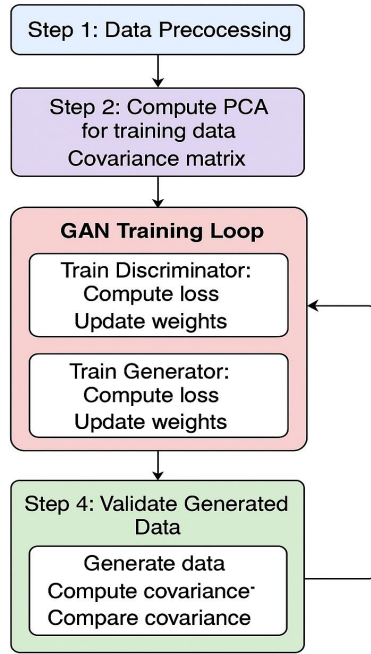


Figure 1 Algorithm Flow

3.3 Data Generation

After the GAN training is completed, we use the GAN with statistical constraints to generate data as the calibration set for the NIR model construction task.

4 Data Experiments

In this paper, we utilize the publicly available pharmaceutical tablet NIR dataset published at the 2002 International Diffuse Reflectance Conference (IDRC), known as the "Shootout" dataset. This dataset consists of spectra from 654 pharmaceutical tablets measured by two spectrometers and is divided into calibration, validation, and test sets. For our experiments, we selected the first 55 samples from Calibration1 as the small-sample dataset and used the corresponding test data for prediction. The experimental results are shown in the figure below.

First, to verify whether the data generated by the statistically constrained GAN used in this paper is consistent with the original small-sample data, the author employs PCA dimensionality reduction, projecting the data onto the two features with the largest variance. By comparing the projection regions, the differences between the generated data and the original data are evaluated.

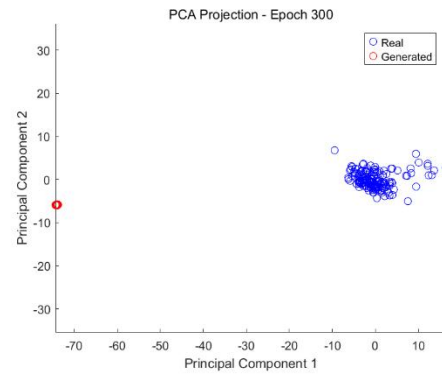


Figure 2 Distribution of Data Generated by the Standard GAN

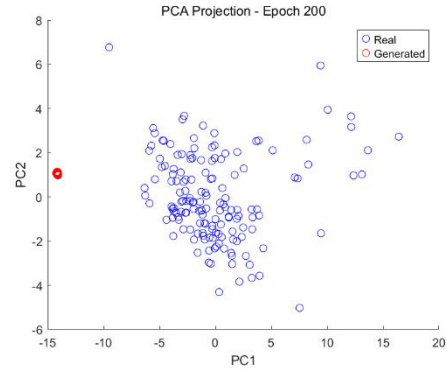


Figure 3 Distribution of Data Generated by the Constrained GAN

Here, the blue data points represent the distribution of real data, and the red data points represent the distribution of generated data. We can observe from Figure 2, under the standard GAN, although training is completed, the generated data significantly differs from the original data in terms of feature distribution. This is largely related to the limitations of small samples. However, from Figure 3, we can see that the distribution of the generated data is very close to that of the original data, indicating that the introduced constraints are effective. This ensures that the generated data not only retains the characteristics of the original data but also expands the dataset.

To evaluate the predictive performance resulting from different generated data, the author selected two common modeling methods, SVR and LSTM, for simulation experiments, thereby demonstrating the positive impact of the data generated by the proposed method on NIR data modeling.

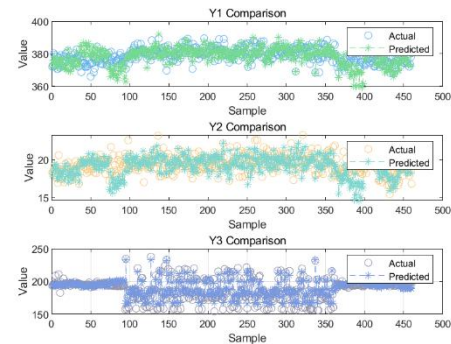


Figure 4 SVR Results Using Data by Constrained GAN

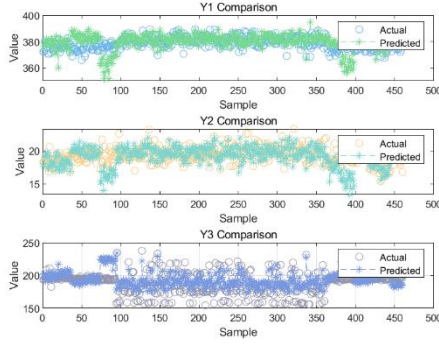


Figure 5 SVR Results Using Data by Standard GAN

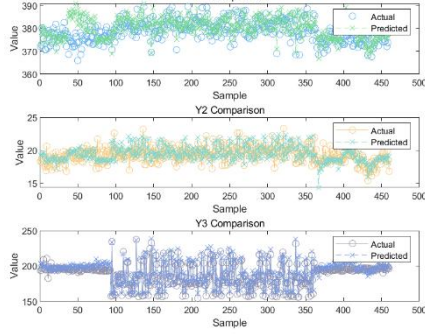


Figure 6 LSTM Results Using Data by Constrained GAN

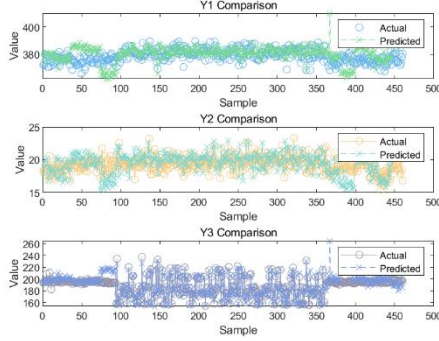


Figure 7 LSTM Results Using Data by Standard GAN

In the experimental results presented, Figure 4 shows the prediction results of the test set using the model built with data generated by the standard GAN, while Figure 5 shows the prediction results using the model built with data generated by the proposed method. From the comparison, we can observe that the model constructed with the proposed method is more accurate and exhibits smaller fluctuations, particularly in predicting samples 50-100 and 350-400. Notably, the prediction of Y3 successfully avoids significant errors. From the statistical metrics shown in Table 1, the RMSE of the proposed method consistently decreases, the RMSE of the prediction results for Y1, Y2, and Y3 decreased by 21.7%, 10.9%, and 54.6%, respectively. This indicates that, under small-sample conditions, if generated samples are used to supplement the calibration set, the higher the similarity between the generated data and the original data, the better the model performance. This also demonstrates that the statistical constraints can provide more accurate and production-relevant constraints compared to the discriminator in GAN. This has significant implications for practical production.

The same results can also be observed in Figure 6 and Figure 7. We find that the LSTM method performs better under the experimental settings, but the model trained with data generated by the proposed method, as shown in Figure 7, exhibits even superior performance. This indicates that the proposed method framework is universally applicable across multiple approaches. Additionally, in Table 2, we also compare the RMSE of the prediction results, the RMSE of the prediction results for Y1, Y2, and Y3 decreased by 13.8%, 34.8%, and 47.2%, respectively. Similarly, the prediction errors for the three key predicted values are reduced due to the introduction of data generated by the proposed method. This further validates the effectiveness of the proposed method and demonstrates its potential for application across various methods.

Table 1: SVR Prediction RMSE Comparison

Target Value	SVR with GAN data	SVR with Constrained GAN data
Y1	6.619	4.8249
Y2	1.0589	0.9423
Y3	13.1134	5.949

Table 2: LSTM Prediction RMSE Comparison

Target Value	LSTM with GAN data	LSTM with Constrained GAN data
Y1	5.6531	4.8728
Y2	1.3418	0.8747
Y3	9.4733	5.0047

5 Conclusion

This paper addresses the issue of insufficient data samples in NIR data modeling by designing a method called "Data Feature-Constrained GAN Data Generation" for generating usable calibration set data that aligns with the characteristics of the detected objects. The method imposes constraints on the generator's cost function by using the variance matrix of the top M principal components from the PCA dimensionality reduction of the original data as a variance constraint for the PCA-reduced generated data. This ensures that the generated data not only deceives the discriminator but also maintains statistical consistency with the existing data. The method was validated on a publicly available dataset.

The results show that the proposed approach, from the perspective of data statistical features, employs a reliable method to constrain the data generation process, thereby enhancing the credibility and information content of the generated data and improving the model's prediction accuracy. This enables more precise measurements of similar items using NIR. The method proposed in this paper has a clear structure and significant potential for engineering applications. Our future work should continue to advance research on its application in miniaturized portable NIR devices and explore model calibration techniques.

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